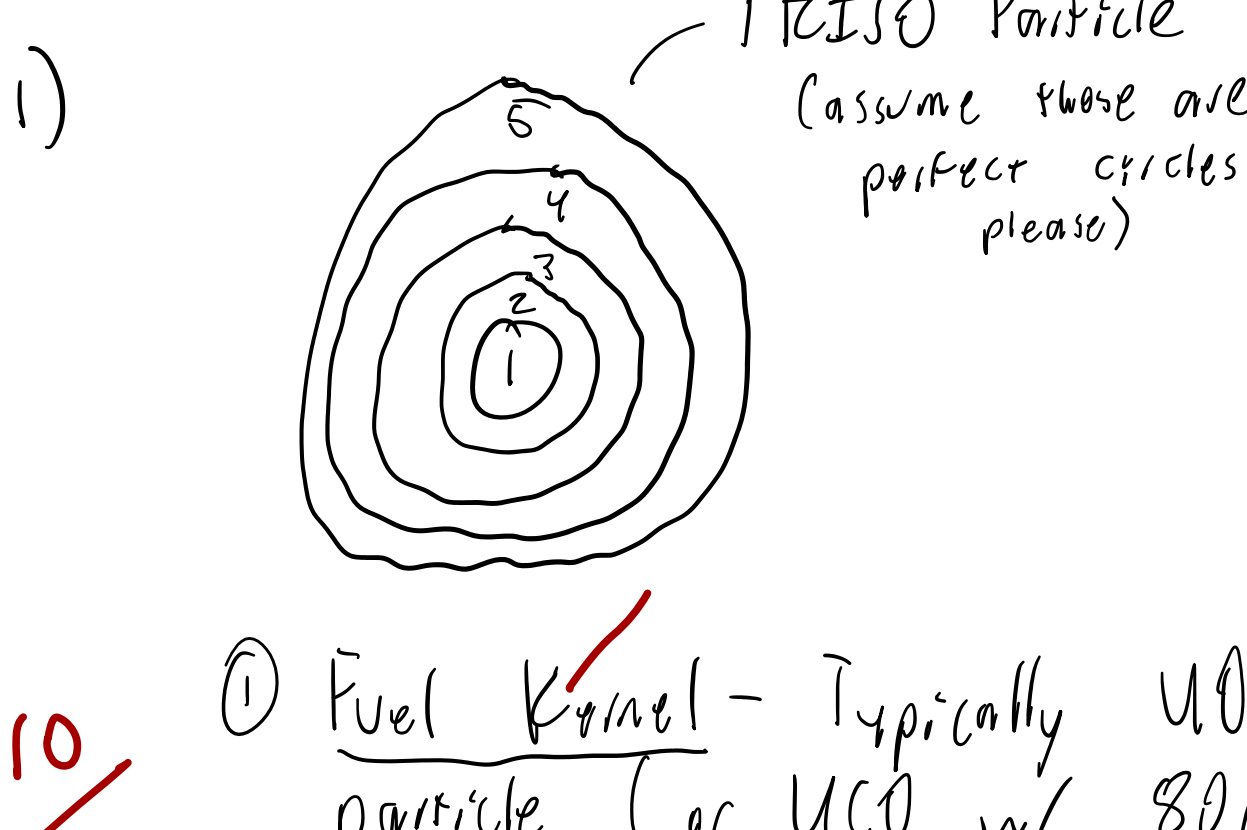


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- ① Fuel Kernel - Typically UO_2 or UC fuel particle (or UCO w/ 80/20 split between Oxide/carbide fuel, respectively), provides fission source/generates heat
- ② Butter - Porous graphite layer that accommodates swelling of the TRISO particle over time and captures fission products (gases)
- ③ IP, C - inner structural layer that protects the SiC layer from fission product attack and ensures SiC layer remains in compression
- ④ SiC - Silicon carbide layer that acts as the primary pressure vessel of the TRISO particle and captures fission products, preventing their release
- ⑤ OP, C - outer structural layer that protects the TRISO particle/SiC layer during manufacturing and ensures the SiC layer remains in compression

2) Oxide fuel kernels are easier to manufacture and have greater fission product retention than carbide fuel. Carbide fuel kernels are able to operate at much higher temperatures while maintaining high thermal efficiency. The primary downside of oxide fuel is the release of free oxygen during the fission process which can form gases with other fission products or the carbon in the graphite butter layer, which can increase the pressure of the particle and possibly cause structural damage through overpressurization. Carbide fuel on the other hand has significantly less fission product retention. Recent TRISO designs have led to an 80/20 split between Oxide/carbide fuel kernels to maintain high fission product retention while minimizing free oxygen levels.

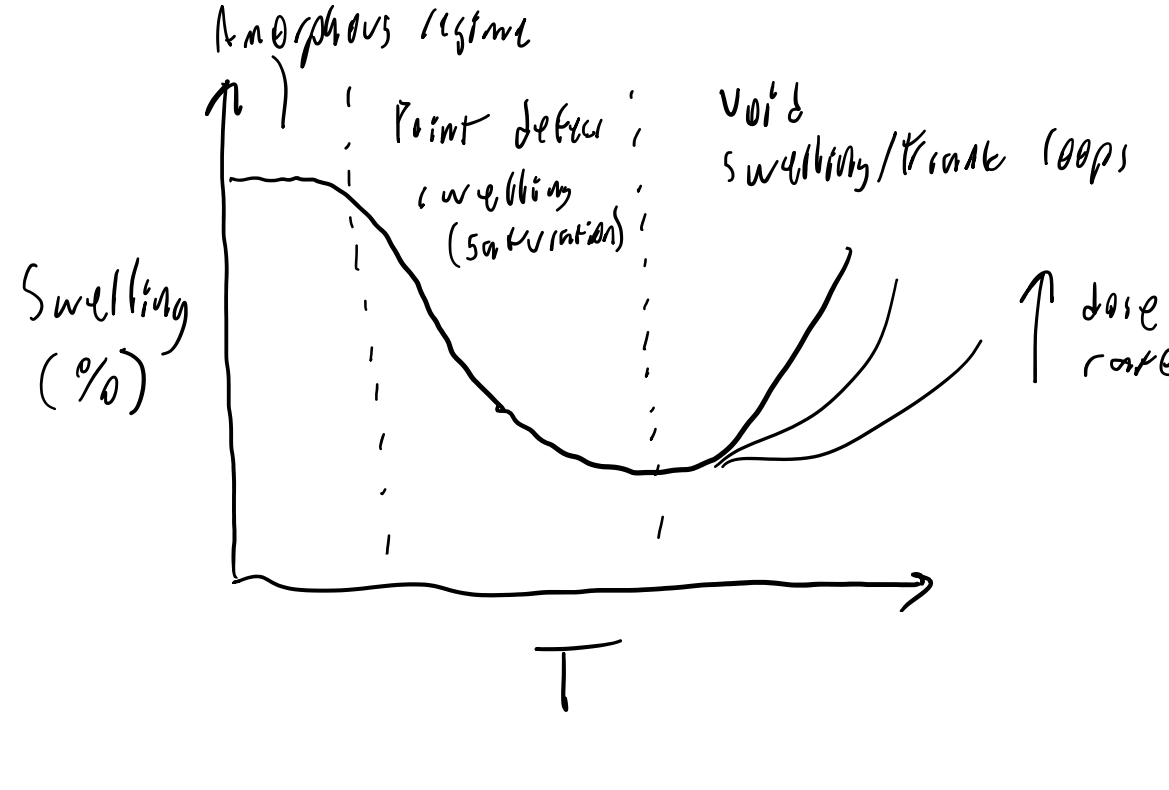
3) A key aspect of fission product chemistry in a TRISO fuel kernel is the production of free oxygen, which can cause a significant buildup of gases (specifically CO) and overpressurize the particle.

The various types of fission products that can form are soluble with insoluble oxides, metals, volatiles, and noble gases. Fission products can escape the fuel particle but diffusing along grain boundaries. The two fission products most able to diffuse along grain boundaries and escape to outer layers are palladium (Pd) and silver (Ag). Pd is the largest concern as it significantly contributes to fission product attack of the SiC layer.

4) There are three temperature regimes of irradiation damage in SiC: a low-T regime ($< 473\text{ K}$), low-med-T regime ($473 - 1300\text{ K}$), and a high-T regime ($1300\text{ K}+$).

- Low-T ($< 473\text{ K}$): 'Black spot' irradiation-induced point defects dominate in this regime. The accumulation of these defects reaches an amorphization point with increasing temperature. Swelling will increase logarithmically until that amorphization point is reached.
- Low-Med-T ($473 - 1300\text{ K}$): This region is characterized by a point defect swelling regime that reaches a saturation point. That saturation point of swelling decreases with increasing temperature (i.e., as temperature increases in this regime, swelling decreases).
- High-T ($1300\text{ K}+$): The high temperature region of irradiation defects sees significant interstitial Frank loops and void swelling defects, which do not saturate. Increasing dose rate and temperature both increase the amount of swelling in this regime.

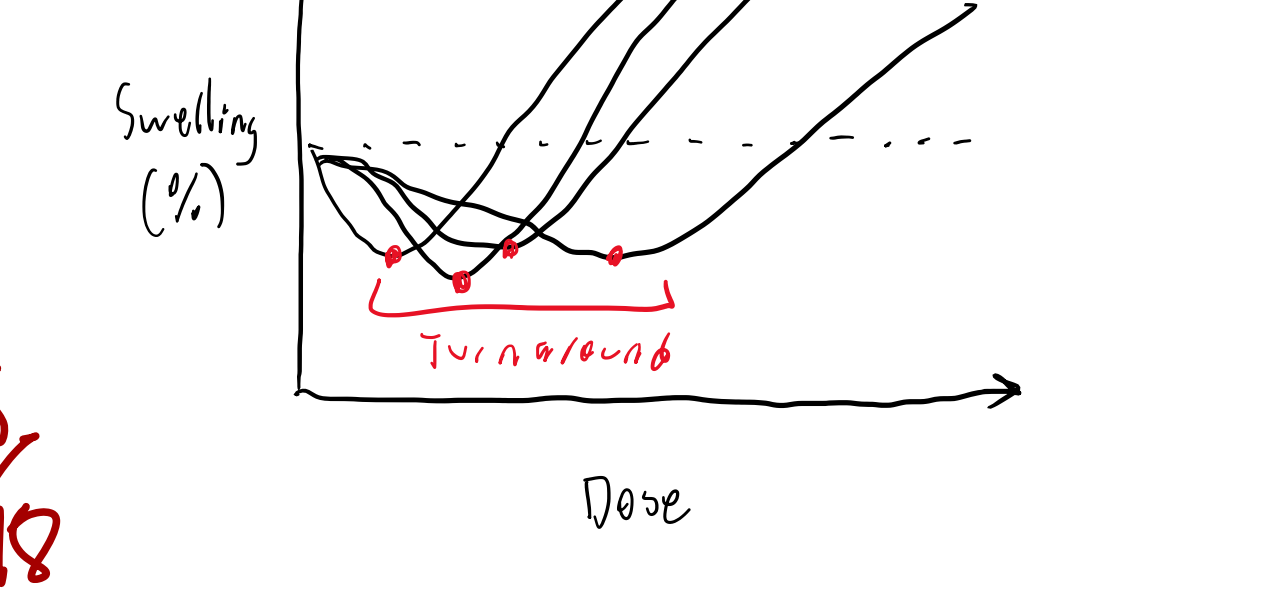
Graphic of Above Discussion



5) Phonon scattering interaction is the primary driver of thermal conductivity changes in SiC with irradiation. At low temperatures, point defect phonon scattering dominates and reaches saturation points at low dose rates. Thermal conductivity will decrease with increasing temperature due to phonon-phonon scattering, and also decrease with increasing dose rate due to the above mentioned phonon-defect scattering.

-voids?

Due to the association of thermal conductivity with point-defect/void swelling of SiC, the amount of swelling can be used as an indirect measurement of thermal conductivity/resistance.



Anisotropic thermal expansion of the graphite layer during production creates a porous layer with cracks/defects that are parallel to the basal plane. As graphite is irradiated, interstitial defects occur between basal planes, which cause an increase in the 'c' direction and a decrease in the 'a' direction. This causes the initial net volume decrease seen in the graph above because the porosity of the graphite can initially accommodate the increase in 'c'. Eventually, when the porosity is filled in and can no longer accommodate the increasing 'c', we see 'turnaround', where the net volume begins to increase with dose rate due to the increase in the 'c' direction. Differing stress states in the graphite layer can affect this behavior, with less shrinkage in compression and more in tension.

7) • Overpressurization: The buildup of fission product gases and oxide gases (specifically CO) can increase pressure inside the TRISO particle and eventually cause a failure of the SiC layer pressure vessel boundary.

• Kernel Migration: Temperature gradients inside the fuel kernel can cause a buildup of carbon on one side of the kernel. Migration of the fuel can eventually cause the kernel to damage/penetrate the outer layers.

• Fission Product Attack: Fission products that escape the fuel kernel via grain boundary diffusion (specifically Pd) can interact with the SiC layer and potentially cause damage/failure.

8) The use of ZrC instead of SiC as the primary pressure vessel layer is being investigated due to the higher temperature stability. ZrC is able to operate at significantly higher temperatures than SiC, making it advantageous for process heat generation use cases of HTGRs. ZrC also has better fission product retention, but is more susceptible to oxidation than SiC is.

9) The concern with a Helium environment when selecting ITH materials is not the He itself but the impurities within the He gas (e.g., CO , O_2 , CO_2 , etc.). These impurities cause a significant corrosion risk for ITH materials. The ITH must balance carburization, decarburization, and oxidation effects. The best way to protect against corrosion/material damage is to have a consistent thin oxide layer on the ITH material. ITH materials must also be able to withstand the very high temperatures seen in HTGRs.